

Group Project - Milestone 1

Background

The shocking truth is ... sabotage !!! (this is the main reason that causes vehicle malfunction during the critical descent into the ... terrestrial environment)

It appears that the IT and Security departments in SpaceXXX had been ... compromised. There was an internal confrontation in which the following happened :

- A fire broke out at multiple levels where all the IT departments were situated
- Many IT team leaders, project managers and employees either died, or suffered serious, terminal injuries
- Important data in servers were either deleted or tampered with
- Widespread destruction (> 90%) of all the critical IT hardware & equipment

From the “scene of crime”, police managed to recover 2 items :

- 1) an un-marked / un-labeled flash drive, containing what appears to be ... programming source code files?
- 2) Some (slightly burnt) hardcopy / colored printouts of what appears to be terminal interactions between user and ... some program

Separate soft copies of the above items have been passed to Agent ‘T’, who is currently attempting to recruit and assemble multiple teams of talented programmers. Exploiting the resources found in the recovered items, all teams are to perform research & analysis, so as to uncover the ‘original’ requirements, and to design, develop + deploy a “controller” system that :

- a) works with the existing programs already installed in the vehicle via satellite communications
- b) helps SpaceXXX regain control of the vehicle, issue commands and getting it to fulfill the requisite missions

The next section describes the set of tasks / instructions for this MS_1 project.

Task Instructions / Info

- A) Based the “raw files” & data from “sample interactions” passed to you by Agent ‘T’, your team have to figure out the answers to several issues :
- (i) What is/are the most important file(s) that your “to-be-developed” system is supposed to integrate/work with?
 - (ii) Are there any sample data / file(s) for your “to-be-developed” system to perform TESTING ON? If yes, how would your system access / load such data?
 - (iii) What are the compilation commands to be issued, so as to generate an executable program incorporating both your source files, together with the ‘raw files’ being passed to you?
 - (iv) Which are the critical functionalities that are already implemented and deployed, that your team can easily access and exploit?
 - (v) What are the processing / functionalities, that your “to-be-developed” system is supposed to implement / provide?
- B) Please refer to **Appendix A**, which provides some administrative instructions as well as a brief description of the raw files include in this MS_1 package.
- C) Please refer to **Appendix B**, which provides some instructions as well screen captures of those hardcopy printouts (recovered by the police), showing some sample program interactions.

APPENDIX A

(The recovered 'raw' files ...)

Included in this MS_1 package, are the following files (recovered by the police)

- SatComRelay.h, SatComRelay.o
- VehController.o, ConfigSettings.o, ExpReport.o, FileUtil.o, RadarSensor.o, Terrain.o, TerrainMap.o
- >= 1 *.dat file

Clue #1 : Preliminary investigations shows that not all files can be accessed / edited by a text editor! This should help you quickly identify and direct your focus on those file(s) & source code which your system should interact closely with ...

Clue #2 : The filenames seemed pretty informative, therefore even if your team cannot directly access most of the files and observe their content, there should be no problem figuring out the purpose of its existence, and how it fits into the overview architecture of SpaceXXX 'original' vehicle controller system.

Warning #1 : Do not, under any circumstance, attempt to "break" or "alter" the binary / hexadecimal contents of those 'inaccessible' files! This will DEFINITELY cause serious issues during program integration and compilation!

Warning #2 : SpaceXXX 'original' vehicle controller system was designed, developed and deployed within Ubuntu (a Linux based) OS. You are to transfer all these raw files DIRECTLY into Ubuntu OS, before making any attempt to develop your system, or to access those files which are ... accessible (i.e. whose contents can be deciphered by Ubuntu text editor).

Warning #3 : Any attempts to open the raw files using text editors on non-Ubuntu OS, will most likely cause problems when these files are subsequently being accessed, compiled and deployed in Ubuntu OS.

APPENDIX B

Screen captures of hardcopy printouts (as recovered by police) ...

Before you start to “dive in” and analyze the screen captures of various program interactions, do consider the following tips / hints, which might shed some light into understanding and interpreting the screen capture’s content!

Hints :

- 1) With reference to the section discussing user interface / interaction in the main project document ‘[yyyy_RSE1201_GroupProject.doc](#)’, can you identify from the screen captures, what kinds of interaction / functionalities should be grouped under :
 - **Stage 1** – e.g. program setup / configuration?
 - **Stage 2** – e.g. main workflow / functionalities / processing?
 - **Stage 3** – e.g. generating / display of summary report(s), program shutdown?
- 2) For each of the stages 1 – 3, analyze the output messages and instructions (from screen captures), work out which messages should come from your “to-be-developed” system, and which are displayed from functions (already implemented) by SpaceXXX
- 3) Are there any relationships between the functionalities offered in the user menu (shown in screen captures), VS the functions you observe from the raw (tip : ‘header’) files described in **Appendix A**?

Even more hints (yaaaay!) :

- A) Are there any relationships between :
 - (i) The type of terrain being scanned, and the amount of energy expended in scanning?
 - (ii) The type of terrain to travel to, and the amount of energy used in shield protection, and movement?
- B) For the same terrain, is the energy consumption amount always the same? Are there situations in which the amount could be different?
- C) What are the consequences of repeatedly :
 - directing vehicle to move to ‘impassable’ / inaccessible terrain?
 - ordering the vehicle to scan a particular direction / grid area which is ‘out of range’?

Riiiiight ... Are yooo READY!? No? Actually I don’t care, ‘cos here comes the screen captures ... !!!

RSE1201 – C Programming

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg
student@Ubuntu-22044-VBox:~/Desktop/RSE1201/ProjectFiles/MS1_Pkg$ gcc demoMainStu.c *.o -o VehExplorer.prog
student@Ubuntu-22044-VBox:~/Desktop/RSE1201/ProjectFiles/MS1_Pkg$ ./VehExplorer.prog

Please enter the scenario file's name      : Scenario05.dat

Is the scenario file encrypted?          (y/n) : y

Do you wish to randomize start position? (y/n) : y

Please select mission type below

1) Casual Explore
2) Find 1 End Point
3) Reach 1 End Point
4) Find ALL End Points
5) Reach ALL End Points
6) Find ALL Danger Points
7) Find Map Boundaries
8) Explore Everything

Enter your choice (1-8) : 5

filename : Scenario05.dat, isFileEncrypted : 1, randomizeStartPosition : 1

Initializing Vehicle Controller ... done!

Initial Enrg : 100000, Tot Enrg Used : 0, Tot Shld Enrg Used : 0, Engy Left : 100000, Shld Enrg Left : 0

*****
*** Welcome to Mr Tian's Vehicle Simulator ! ***
*****

[ Usage Instructions ]
-----

Enter 'q' to Quit

Enter 'v' to show this Usage Instruction (again ..)

'w', 's', 'a', 'd' keys to MOVE vehicle North, South, West & East

Enable 'Numlock', (right key pad), keys 1-9 (except 5) to SCAN terrain :
SW (1), S (2), SE (3), W (4), E (6), NW (7), N (8), NE (9) respectively

't' key = Toggle Scan Radius (betw. 1-2)

'i' key = increase / divert energy to shield

'z' key = Info on Vehicle Status
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg
```

```
'z' key = Info on Vehicle Status  
'c' key = Map of terrain scanned by Vehicle  
'x' key = Map of terrain travelled by Vehicle  
  
Note : In map display, '_' underline indicates curr. veh. location!  
-----  
Press 'y' key and hit Enter to continue! y  
  
Enter your command (q to quit) : z  
Vehicle Status / Statistics :  
Initial Enrg : 100000, Tot Enrg Used : 0, Tot Shld Enrg Used : 0, Engy Left : 100000, Shld Enrg Left : 0  
  
Enter your command (q to quit) : x  
Currently Explored Terrain Layout :  
  
-----  
-- (TUTOR Ver) Print Explored Map! --  
-----  
  
_
```

```
Enter your command (q to quit) : c  
Currently Scanned Terrain Layout :  
  
-----  
-- (TUTOR Ver) Print Scanned Map! --  
-----  
  
_
```

```
Enter your command (q to quit) : z
```

RSE1201 – C Programming

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

Enter your command (q to quit) : z
Vehicle Status / Statistics :
Initial Enrg : 100000, Tot Enrg Used : 0, Tot Shld Enrg Used : 0, Engy Left : 100000, Shld Enrg Left : 0

Enter your command (q to quit) : i
Enter amt. of energy to increase / allocate to shields (1 - 100000) : 100001
Error! Insufficient energy reserves. Try reducing the quantity allocated to shield!

Enter your command (q to quit) : z
Vehicle Status / Statistics :
Initial Enrg : 100000, Tot Enrg Used : 0, Tot Shld Enrg Used : 0, Engy Left : 100000, Shld Enrg Left : 0

Enter your command (q to quit) : i
Enter amt. of energy to increase / allocate to shields (1 - 100000) : -5
Error! Invalid energy qty '-5' for shield allocation!

Enter your command (q to quit) : z
Vehicle Status / Statistics :
Initial Enrg : 100000, Tot Enrg Used : 0, Tot Shld Enrg Used : 0, Engy Left : 100000, Shld Enrg Left : 0

Enter your command (q to quit) : i
Enter amt. of energy to increase / allocate to shields (1 - 100000) : 20000
20000 energy units allocated to shields!

Enter your command (q to quit) : z
Vehicle Status / Statistics :
Initial Enrg : 100000, Tot Enrg Used : 20000, Tot Shld Enrg Used : 0, Engy Left : 80000, Shld Enrg Left : 20000

Enter your command (q to quit) : t
Toggle Scan Radius => now set to 2 terrain units!

Enter your command (q to quit) : t
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

Toggle Scan Radius => now set to 2 terrain units!

Enter your command (q to quit) : t
Toggle Scan Radius => now set to 1 terrain units!

Enter your command (q to quit) : 1

Returned (scanned) terrain symbol is : 'h'
Initial Enrg : 100000, Tot Enrg Used : 20360, Tot Shld Enrg Used : 0, Engy Left : 79640, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      h  -

Enter your command (q to quit) : 2

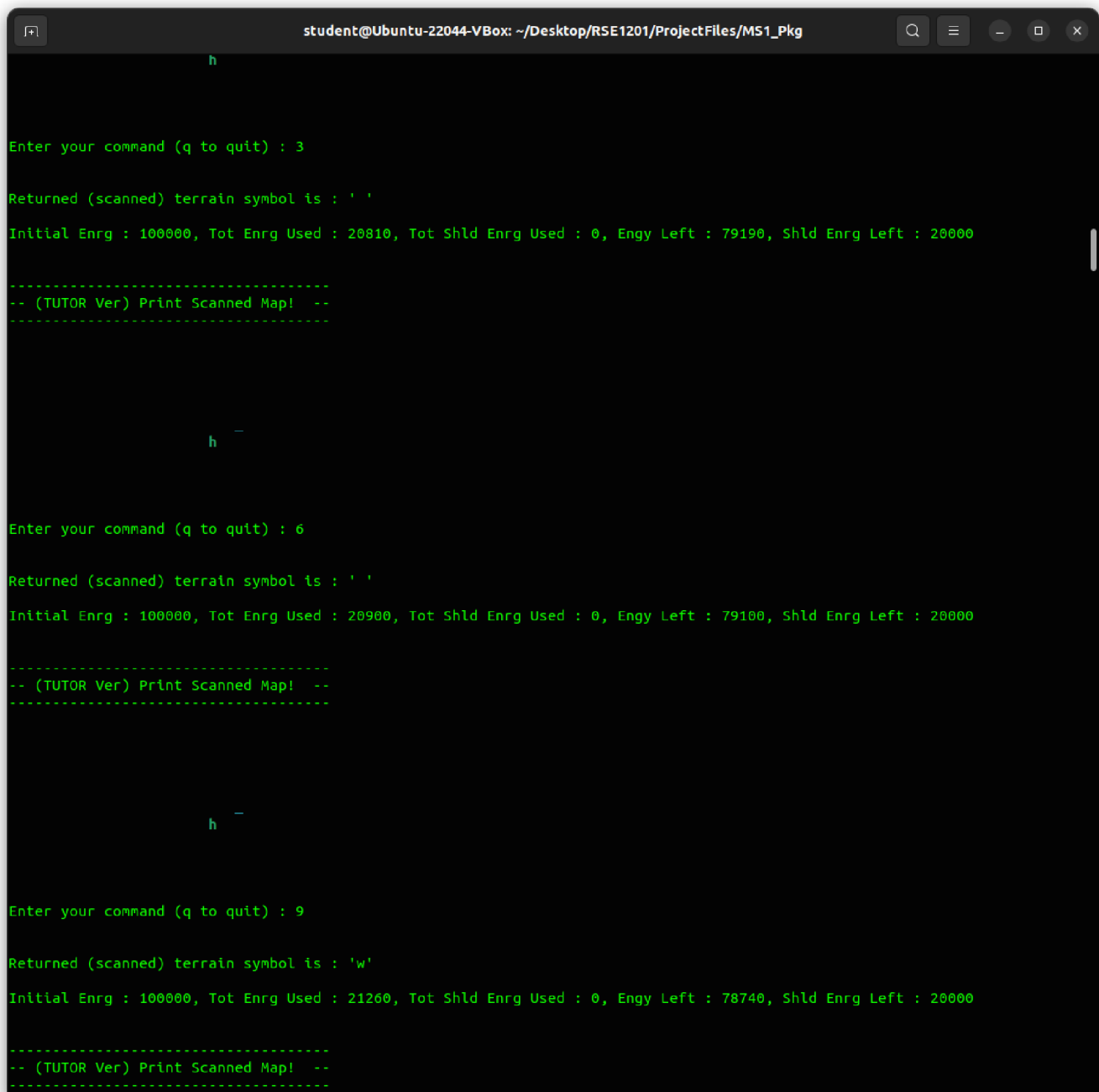
Returned (scanned) terrain symbol is : ' '
Initial Enrg : 100000, Tot Enrg Used : 20450, Tot Shld Enrg Used : 0, Engy Left : 79550, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      h  -

Enter your command (q to quit) : 3

Returned (scanned) terrain symbol is : ' '
Initial Enrg : 100000, Tot Enrg Used : 20810, Tot Shld Enrg Used : 0, Engy Left : 79190, Shld Enrg Left : 20000
```



```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

h

Enter your command (q to quit) : 3

Returned (scanned) terrain symbol is : ' '
Initial Enrg : 100000, Tot Enrg Used : 20810, Tot Shld Enrg Used : 0, Engy Left : 79190, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

h

Enter your command (q to quit) : 6

Returned (scanned) terrain symbol is : ' '
Initial Enrg : 100000, Tot Enrg Used : 20900, Tot Shld Enrg Used : 0, Engy Left : 79100, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

h

Enter your command (q to quit) : 9

Returned (scanned) terrain symbol is : 'w'
Initial Enrg : 100000, Tot Enrg Used : 21260, Tot Shld Enrg Used : 0, Engy Left : 78740, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg
Initial Enrg : 100000, Tot Enrg Used : 21260, Tot Shld Enrg Used : 0, Engy Left : 78740, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      w
      -
h

Enter your command (q to quit) : 8

Returned (scanned) terrain symbol is : 'M'

Initial Enrg : 100000, Tot Enrg Used : 21350, Tot Shld Enrg Used : 0, Engy Left : 78650, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      M w
      -
h

Enter your command (q to quit) : 7

Returned (scanned) terrain symbol is : ' '

Initial Enrg : 100000, Tot Enrg Used : 21710, Tot Shld Enrg Used : 0, Engy Left : 78290, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      M w
      -
h
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      M  w
      |
h    -

Enter your command (q to quit) : 4

Returned (scanned) terrain symbol is : ' '
Initial Enrg : 100000, Tot Enrg Used : 21800, Tot Shld Enrg Used : 0, Engy Left : 78200, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      M  w
      |
h    -

Enter your command (q to quit) : 5

Invalid Command : '5', please try again!

Enter your command (q to quit) : t

Toggle Scan Radius => now set to 2 terrain units!

Enter your command (q to quit) : 1

Returned (scanned) terrain symbol is : '#'
Initial Enrg : 100000, Tot Enrg Used : 22520, Tot Shld Enrg Used : 0, Engy Left : 77480, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg
Initial Enrg : 100000, Tot Enrg Used : 22520, Tot Shld Enrg Used : 0, Enrg Left : 77480, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      M W
      |
h     |
#     |

Enter your command (q to quit) : 2

Returned (scanned) terrain symbol is : '#'

Initial Enrg : 100000, Tot Enrg Used : 22700, Tot Shld Enrg Used : 0, Enrg Left : 77300, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      M W
      |
h     |
#     |
#     |

Enter your command (q to quit) : 3

Returned (scanned) terrain symbol is : '#'

Initial Enrg : 100000, Tot Enrg Used : 23420, Tot Shld Enrg Used : 0, Enrg Left : 76580, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      M W
      |
h     |
#     |
#     |
#     |
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-- (TUTOR Ver) Print Scanned Map! --
-----
      M  W
      |
h    |
#    |    #
-----

Enter your command (q to quit) : 6

Returned (scanned) terrain symbol is : '~'

Initial Enrg : 100000, Tot Enrg Used : 23600, Tot Shld Enrg Used : 0, Engy Left : 76400, Shld Enrg Left : 20000

-- (TUTOR Ver) Print Scanned Map! --
-----
      M  W
      |  ~
h    |
#    |    #
-----

Enter your command (q to quit) : 9

Returned (scanned) terrain symbol is : ' '

Initial Enrg : 100000, Tot Enrg Used : 24320, Tot Shld Enrg Used : 0, Engy Left : 75680, Shld Enrg Left : 20000

-- (TUTOR Ver) Print Scanned Map! --
-----
      M  W
      |  ~
h    |
#    |    #
-----

Enter your command (q to quit) : 8
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg
Initial Enrg : 100000, Tot Enrg Used : 24320, Tot Shld Enrg Used : 0, Engy Left : 75680, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      M  W
      -  ~
h      #  #
#      #  #

Enter your command (q to quit) : 8

Returned (scanned) terrain symbol is : 'M'

Initial Enrg : 100000, Tot Enrg Used : 24500, Tot Shld Enrg Used : 0, Engy Left : 75500, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      M
      M  W
      -  ~
h      #  #
#      #  #

Enter your command (q to quit) : 7

Returned (scanned) terrain symbol is : ' '

Initial Enrg : 100000, Tot Enrg Used : 25220, Tot Shld Enrg Used : 0, Engy Left : 74780, Shld Enrg Left : 20000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      M
      M  W
      -  ~
h      #  #
#      #  #
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-- (TUTOR Ver) Print Scanned Map! --
-----
      M
      M W
      - ~
h    -
#    #    #

Enter your command (q to quit) : 4

Returned (scanned) terrain symbol is : ' '

Initial Enrg : 100000, Tot Enrg Used : 25400, Tot Shld Enrg Used : 0, Engy Left : 74600, Shld Enrg Left : 20000

-- (TUTOR Ver) Print Scanned Map! --
-----
      M
      M W
      - ~
h    -
#    #    #

Enter your command (q to quit) : z

Vehicle Status / Statistics :

Initial Enrg : 100000, Tot Enrg Used : 25400, Tot Shld Enrg Used : 0, Engy Left : 74600, Shld Enrg Left : 20000

Enter your command (q to quit) : c

Currently Scanned Terrain Layout :

-- (TUTOR Ver) Print Scanned Map! --
-----
      M
      M W
      - ~
h    -
#    #    #
```

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```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-- (TUTOR Ver) Print Scanned Map! --
-----
      M
      M W
      - ~
h    -
#    #    #

Enter your command (q to quit) : x
Currently Explored Terrain Layout :

-----
-- (TUTOR Ver) Print Explored Map! --
-----

      -

Enter your command (q to quit) : w

Vehicle moved 'Up / North' by 1 terrain unit.
Initial Enrg : 100000, Tot Enrg Used : 26200, Tot Shld Enrg Used : 100, Enrg Left : 73800, Shld Enrg Left : 19900

-----
-- (TUTOR Ver) Print Explored Map! --
-----

      M

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.
Initial Enrg : 100000, Tot Enrg Used : 26300, Tot Shld Enrg Used : 100, Enrg Left : 73700, Shld Enrg Left : 19900
```

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```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.
Initial Enrg : 100000, Tot Enrg Used : 26300, Tot Shld Enrg Used : 100, Engy Left : 73700, Shld Enrg Left : 19900

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  _ M

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.
Initial Enrg : 100000, Tot Enrg Used : 26400, Tot Shld Enrg Used : 100, Engy Left : 73600, Shld Enrg Left : 19900

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  _ ■ M

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.
Initial Enrg : 100000, Tot Enrg Used : 27200, Tot Shld Enrg Used : 200, Engy Left : 72800, Shld Enrg Left : 19800

-----
-- (TUTOR Ver) Print Explored Map! --
-----

M ■ ■ M
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg
Vehicle moved 'Left / West' by 1 terrain unit.
Initial Enrg : 100000, Tot Enrg Used : 27200, Tot Shld Enrg Used : 200, Engy Left : 72800, Shld Enrg Left : 19800

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  M  █  █  M

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.
Initial Enrg : 100000, Tot Enrg Used : 28200, Tot Shld Enrg Used : 1200, Engy Left : 71800, Shld Enrg Left : 18800

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  E  M  █  █  M

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.
Initial Enrg : 100000, Tot Enrg Used : 28400, Tot Shld Enrg Used : 1250, Engy Left : 71600, Shld Enrg Left : 18750

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  f  E  M  █  █  M
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

Vehicle moved 'Left / West' by 1 terrain unit.
Initial Enrg : 100000, Tot Enrg Used : 28400, Tot Shld Enrg Used : 1250, Engy Left : 71600, Shld Enrg Left : 18750

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  f  E  M  |  |  M

Enter your command (q to quit) : 1

Returned (scanned) terrain symbol is : '#'
Initial Enrg : 100000, Tot Enrg Used : 29120, Tot Shld Enrg Used : 1250, Engy Left : 70880, Shld Enrg Left : 18750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      _           M
      M           M  W
#           h           ~
#           #           #

Enter your command (q to quit) : 4

Returned (scanned) terrain symbol is : '#'
Initial Enrg : 100000, Tot Enrg Used : 29300, Tot Shld Enrg Used : 1250, Engy Left : 70700, Shld Enrg Left : 18750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#           _           M
#           M           M  W
#           h           ~
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#      _      M
      M w
#      h      ~
      #  #  #

Enter your command (q to quit) : t
Toggle Scan Radius => now set to 1 terrain units!

Enter your command (q to quit) : 1

Returned (scanned) terrain symbol is : ' '
Initial Enrg : 100000, Tot Enrg Used : 29660, Tot Shld Enrg Used : 1250, Enrg Left : 70340, Shld Enrg Left : 18750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#      _      M
      M w
#      h      ~
      #  #  #

Enter your command (q to quit) : 4

Returned (scanned) terrain symbol is : ' '
Initial Enrg : 100000, Tot Enrg Used : 29750, Tot Shld Enrg Used : 1250, Enrg Left : 70250, Shld Enrg Left : 18750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#      _      M
      M w
```

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```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

#           h           ~
#           #           #

Enter your command (q to quit) : 7

Returned (scanned) terrain symbol is : ' '

Initial Enrg : 100000, Tot Enrg Used : 30110, Tot Shld Enrg Used : 1250, Engy Left : 69890, Shld Enrg Left : 18750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#           _           M           ~
#           #           #           #
#           h           #           #

Enter your command (q to quit) : 8

Returned (scanned) terrain symbol is : 'X'

Initial Enrg : 100000, Tot Enrg Used : 30200, Tot Shld Enrg Used : 1250, Engy Left : 69800, Shld Enrg Left : 18750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#           X           _           M           ~
#           #           #           #
#           h           #           #

Enter your command (q to quit) : 9

Returned (scanned) terrain symbol is : 'f'

Initial Enrg : 100000, Tot Enrg Used : 30560, Tot Shld Enrg Used : 1250, Engy Left : 69440, Shld Enrg Left : 18750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

Initial Enrg : 100000, Tot Enrg Used : 30560, Tot Shld Enrg Used : 1250, Engy Left : 69440, Shld Enrg Left : 18750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#      X  f      M
#      _      M  w
#              h      ~
#              #  #  #

Enter your command (q to quit) : 6

Returned (scanned) terrain symbol is : 'E'

Initial Enrg : 100000, Tot Enrg Used : 30650, Tot Shld Enrg Used : 1250, Engy Left : 69350, Shld Enrg Left : 18750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#      X  f      M
#      _  E      M  w
#              h      ~
#              #  #  #

Enter your command (q to quit) : 3

Returned (scanned) terrain symbol is : ' '

Initial Enrg : 100000, Tot Enrg Used : 31010, Tot Shld Enrg Used : 1250, Engy Left : 68990, Shld Enrg Left : 18750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#      X  f      M
#      _  E      M  w
#              h      ~
#              #  #  #
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-- (TUTOR Ver) Print Scanned Map! --
-----
#      X f      M
#      _ E      M w
#              h
#              # # #

Enter your command (q to quit) : 2

Returned (scanned) terrain symbol is : 'X'

Initial Enrg : 100000, Tot Enrg Used : 31100, Tot Shld Enrg Used : 1250, Enrg Left : 68900, Shld Enrg Left : 18750

-- (TUTOR Ver) Print Scanned Map! --
-----
#      X f      M
#      _ E      M w
#      X
#              h
#              # # #

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.

Initial Enrg : 100000, Tot Enrg Used : 31200, Tot Shld Enrg Used : 1250, Enrg Left : 68800, Shld Enrg Left : 18750

-- (TUTOR Ver) Print Explored Map! --
-----

_ f E M ■ ■ M

Enter your command (q to quit) : s
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

Enter your command (q to quit) : s

Vehicle moved 'Down / South' by 1 terrain unit.
Initial Enrg : 100000, Tot Enrg Used : 31300, Tot Shld Enrg Used : 1250, Engy Left : 68700, Shld Enrg Left : 18750

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  f E M  M
  -

Enter your command (q to quit) : s

Vehicle moved 'Down / South' by 1 terrain unit.
Initial Enrg : 100000, Tot Enrg Used : 32300, Tot Shld Enrg Used : 2250, Engy Left : 67700, Shld Enrg Left : 17750

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  f E M  M
  E

Enter your command (q to quit) : s

INVALID / IMPASSABLE LOCATION! Unable to move SOUTH!
Initial Enrg : 100000, Tot Enrg Used : 33300, Tot Shld Enrg Used : 7250, Engy Left : 66700, Shld Enrg Left : 12750

-----
-- (TUTOR Ver) Print Explored Map! --
-----
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  f E M  M
  E

Enter your command (q to quit) : 1

Returned (scanned) terrain symbol is : '#'
Initial Enrg : 100000, Tot Enrg Used : 33660, Tot Shld Enrg Used : 7250, Engy Left : 66340, Shld Enrg Left : 12750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#   X f       M
#   E       M w
#   X
#   h
#   #   #

Enter your command (q to quit) : 2

Returned (scanned) terrain symbol is : '#'
Initial Enrg : 100000, Tot Enrg Used : 33750, Tot Shld Enrg Used : 7250, Engy Left : 66250, Shld Enrg Left : 12750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#   X f       M
#   E       M w
#   X
#   h
#   #   #
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-- (TUTOR Ver) Print Scanned Map! --
-----
#      X f      M
#      E      M w
#      X      ~
#      h
# #      # # #

Enter your command (q to quit) : 3

Returned (scanned) terrain symbol is : '#'

Initial Enrg : 100000, Tot Enrg Used : 34110, Tot Shld Enrg Used : 7250, Engy Left : 65890, Shld Enrg Left : 12750

-- (TUTOR Ver) Print Scanned Map! --
-----
#      X f      M
#      E      M w
#      X      ~
#      h
# #      # # #

Enter your command (q to quit) : 6

Returned (scanned) terrain symbol is : ' '

Initial Enrg : 100000, Tot Enrg Used : 34200, Tot Shld Enrg Used : 7250, Engy Left : 65800, Shld Enrg Left : 12750

-- (TUTOR Ver) Print Scanned Map! --
-----
#      X f      M
#      E      M w
#      X      ~
#      h
# #      # # #

Enter your command (q to quit) : 8
```

RSE1201 – C Programming

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

Initial Enrg : 100000, Tot Enrg Used : 34200, Tot Shld Enrg Used : 7250, Engy Left : 65800, Shld Enrg Left : 12750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#      X f      M
#      E      M w
#      X      ~
# _      h
# # #      # # #

Enter your command (q to quit) : 8

Returned (scanned) terrain symbol is : ' '

Initial Enrg : 100000, Tot Enrg Used : 34290, Tot Shld Enrg Used : 7250, Engy Left : 65710, Shld Enrg Left : 12750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#      X f      M
#      E      M w
#      X      ~
# _      h
# # #      # # #

Enter your command (q to quit) : 7

Returned (scanned) terrain symbol is : '#'

Initial Enrg : 100000, Tot Enrg Used : 34650, Tot Shld Enrg Used : 7250, Engy Left : 65350, Shld Enrg Left : 12750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#      X f      M
#      E      M w
#      X      ~
# _      h
# # #      # # #
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

Returned (scanned) terrain symbol is : '#'

Initial Enrg : 100000, Tot Enrg Used : 34650, Tot Shld Enrg Used : 7250, Engy Left : 65350, Shld Enrg Left : 12750

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#      X f      M
#      E      M W
#      X      ~
#      _      h
# # #      # # #

Enter your command (q to quit) : z
Vehicle Status / Statistics :
Initial Enrg : 100000, Tot Enrg Used : 34650, Tot Shld Enrg Used : 7250, Engy Left : 65350, Shld Enrg Left : 12750

Enter your command (q to quit) : x
Currently Explored Terrain Layout :

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  f E M  M
  E

Enter your command (q to quit) : s

INVALID / IMPASSABLE LOCATION! Unable to move SOUTH!

Initial Enrg : 100000, Tot Enrg Used : 35650, Tot Shld Enrg Used : 12250, Engy Left : 64350, Shld Enrg Left : 7750

-----
-- (TUTOR Ver) Print Explored Map! --
-----
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

INVALID / IMPASSABLE LOCATION! Unable to move SOUTH!
Initial Enrg : 100000, Tot Enrg Used : 35650, Tot Shld Enrg Used : 12250, Engy Left : 64350, Shld Enrg Left : 7750

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  f E M  M
E

Enter your command (q to quit) : d

Vehicle moved 'Right / East' by 1 terrain unit.
Initial Enrg : 100000, Tot Enrg Used : 35750, Tot Shld Enrg Used : 12250, Engy Left : 64250, Shld Enrg Left : 7750

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  f E M  M
E _

Enter your command (q to quit) : s

INVALID / IMPASSABLE LOCATION! Unable to move SOUTH!
Initial Enrg : 100000, Tot Enrg Used : 36750, Tot Shld Enrg Used : 17250, Engy Left : 63250, Shld Enrg Left : 2750

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  f E M  M
E _
```

RSE1201 – C Programming

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

Enter your command (q to quit) : s

INVALID / IMPASSABLE LOCATION! Unable to move SOUTH!

Initial Enrg : 100000, Tot Enrg Used : 36750, Tot Shld Enrg Used : 17250, Engy Left : 63250, Shld Enrg Left : 2750

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  f E M  M
E _

Enter your command (q to quit) : s

INVALID / IMPASSABLE LOCATION! Unable to move SOUTH!

Initial Enrg : 100000, Tot Enrg Used : 37750, Tot Shld Enrg Used : 20000, Engy Left : 62250, Shld Enrg Left : 0

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  f E M  M
E _

Enter your command (q to quit) : s

The shields are down, vehicle is DESTROYED! (curr shld enrgy : 0)

*** MISSION OVER!! ***

[ Mission Statistics ]

Mission Type      : Find All End Points
End Points found  : 2
Completion Status : 100.00 %

SAFELY DEALLOCATE MEMORY !!
student@Ubuntu-22044-VBox:~/Desktop/RSE1201/ProjectFiles/MS1_Pkg$ -|
```

Well ... hope the above screen captures prove useful for your team to analyze, in terms of the expected program behavior, the functionalities to be implemented, and ...

What? You want more? You sure about this?

Be careful what you wish for ...

Alright, the next series of screen captures would be EVERYTHING that Agent 'T' have.

See if you can discover / learn something new, that was not found in the earlier series of screen caps ...

Here comes the 2nd set !!

RSE1201 – C Programming

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

SAFELY DEALLOCATE MEMORY !!
student@Ubuntu-22044-VBox:~/Desktop/RSE1201/ProjectFiles/MS1_Pkg$ ./VehExplorer.prog

Please enter the scenario file's name      : Scenario05.dat

Is the scenario file encrypted?          (y/n) : y

Do you wish to randomize start position? (y/n) : y

Please select mission type below

1) Casual Explore
2) Find 1 End Point
3) Reach 1 End Point
4) Find ALL End Points
5) Reach ALL End Points
6) Find ALL Danger Points
7) Find Map Boundaries
8) Explore Everything

Enter your choice (1-8) : 6

filename : Scenario05.dat, isFileEncrypted : 1, randomizeStartPosition : 1
Initializing Vehicle Controller ... done!
Initial Enrg : 120000, Tot Enrg Used : 0, Tot Shld Enrg Used : 0, Enrg Left : 120000, Shld Enrg Left : 0

*****
*** Welcome to Mr Tian's Vehicle Simulator ! ***
*****

[ Usage Instructions ]
-----
Enter 'q' to Quit
Enter 'v' to show this Usage Instruction (again ..)
'w', 's', 'a', 'd' keys to MOVE vehicle North, South, West & East
Enable 'Numlock', (right key pad), keys 1-9 (except 5) to SCAN terrain :
SW (1), S (2), SE (3), W (4), E (6), NW (7), N (8), NE (9) respectively
't' key = Toggle Scan Radius (betw. 1-2)
'i' key = increase / divert energy to shield
'z' key = Info on Vehicle Status
```

RSE1201 – C Programming

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

'i' key = increase / divert energy to shield
'z' key = Info on Vehicle Status
'c' key = Map of terrain scanned by Vehicle
'x' key = Map of terrain travelled by Vehicle

Note : In map display, '_' underline indicates curr. veh. location!
-----

Press 'y' key and hit Enter to continue! y

Enter your command (q to quit) : z
Vehicle Status / Statistics :
Initial Enrg : 120000, Tot Enrg Used : 0, Tot Shld Enrg Used : 0, Engy Left : 120000, Shld Enrg Left : 0

Enter your command (q to quit) : i
Enter amt. of energy to increase / allocate to shields (1 - 120000) : 30000
30000 energy units allocated to shields!

Enter your command (q to quit) : z
Vehicle Status / Statistics :
Initial Enrg : 120000, Tot Enrg Used : 30000, Tot Shld Enrg Used : 0, Engy Left : 90000, Shld Enrg Left : 30000

Enter your command (q to quit) : 1

Returned (scanned) terrain symbol is : ' '
Initial Enrg : 120000, Tot Enrg Used : 30360, Tot Shld Enrg Used : 0, Engy Left : 89640, Shld Enrg Left : 30000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

Enter your command (q to quit) : 2
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

Enter your command (q to quit) : 2

Returned (scanned) terrain symbol is : 'M'

Initial Enrg : 120000, Tot Enrg Used : 30450, Tot Shld Enrg Used : 0, Engy Left : 89550, Shld Enrg Left : 30000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      M

Enter your command (q to quit) : 3

Returned (scanned) terrain symbol is : ' '

Initial Enrg : 120000, Tot Enrg Used : 30810, Tot Shld Enrg Used : 0, Engy Left : 89190, Shld Enrg Left : 30000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      M

Enter your command (q to quit) : 6

Returned (scanned) terrain symbol is : ' '

Initial Enrg : 120000, Tot Enrg Used : 30900, Tot Shld Enrg Used : 0, Engy Left : 89100, Shld Enrg Left : 30000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

  ~
  M

Enter your command (q to quit) : 9

Returned (scanned) terrain symbol is : ' '

Initial Enrg : 120000, Tot Enrg Used : 31260, Tot Shld Enrg Used : 0, Engy Left : 88740, Shld Enrg Left : 30000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

  ~
  M

Enter your command (q to quit) : 8

Returned (scanned) terrain symbol is : '~'

Initial Enrg : 120000, Tot Enrg Used : 31350, Tot Shld Enrg Used : 0, Engy Left : 88650, Shld Enrg Left : 30000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

  ~
  ~
  M

Enter your command (q to quit) : 7
```

RSE1201 – C Programming

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

Enter your command (q to quit) : 7

Returned (scanned) terrain symbol is : '~'

Initial Enrg : 120000, Tot Enrg Used : 31710, Tot Shld Enrg Used : 0, Engy Left : 88290, Shld Enrg Left : 30000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      ~ ~
      -
      M

Enter your command (q to quit) : 4

Returned (scanned) terrain symbol is : 'M'

Initial Enrg : 120000, Tot Enrg Used : 31800, Tot Shld Enrg Used : 0, Engy Left : 88200, Shld Enrg Left : 30000

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      ~ ~
      M -
      M

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.

Initial Enrg : 120000, Tot Enrg Used : 32600, Tot Shld Enrg Used : 100, Engy Left : 87400, Shld Enrg Left : 29900

-----
-- (TUTOR Ver) Print Explored Map! --
-----

      M
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-- (TUTOR Ver) Print Scanned Map! --
-----
      ~ ~
      M
      M
      ~
  #  #

Enter your command (q to quit) : 3

Returned (scanned) terrain symbol is : '#'

Initial Enrg : 120000, Tot Enrg Used : 33610, Tot Shld Enrg Used : 100, Engy Left : 86390, Shld Enrg Left : 29900

-- (TUTOR Ver) Print Scanned Map! --
-----
      ~ ~
      M
      M
      ~
  #  #

Enter your command (q to quit) : t

Toggle Scan Radius => now set to 2 terrain units!

Enter your command (q to quit) : 1

Error! Scanning location is BEYOND map range!

Returned (scanned) terrain symbol is : ''

Initial Enrg : 120000, Tot Enrg Used : 34330, Tot Shld Enrg Used : 100, Engy Left : 85670, Shld Enrg Left : 29900

-- (TUTOR Ver) Print Scanned Map! --
-----
      ~ ~
```

RSE1201 – C Programming

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-----
-- (TUTOR Ver) Print Explored Map! --
-----

      M
      |
      -

Enter your command (q to quit) : 1

Returned (scanned) terrain symbol is : '#'

Initial Enrg : 120000, Tot Enrg Used : 33160, Tot Shld Enrg Used : 100, Engy Left : 86840, Shld Enrg Left : 29900

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      ~ ~
      M  M
      |  |
      -  -
      #  #

Enter your command (q to quit) : 2

Returned (scanned) terrain symbol is : '#'

Initial Enrg : 120000, Tot Enrg Used : 33250, Tot Shld Enrg Used : 100, Engy Left : 86750, Shld Enrg Left : 29900

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      ~ ~
      M  M
      |  |
      -  -
      #  #

Enter your command (q to quit) : 3
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-- (TUTOR Ver) Print Scanned Map! --
-----
      ~ ~
      M
      M
      # #
-----

Enter your command (q to quit) : 3

Returned (scanned) terrain symbol is : '#'

Initial Enrg : 120000, Tot Enrg Used : 33610, Tot Shld Enrg Used : 100, Engy Left : 86390, Shld Enrg Left : 29900

-- (TUTOR Ver) Print Scanned Map! --
-----
      ~ ~
      M
      M
      # #
-----

Enter your command (q to quit) : t

Toggle Scan Radius => now set to 2 terrain units!

Enter your command (q to quit) : 1

Error! Scanning location is BEYOND map range!

Returned (scanned) terrain symbol is : ''

Initial Enrg : 120000, Tot Enrg Used : 34330, Tot Shld Enrg Used : 100, Engy Left : 85670, Shld Enrg Left : 29900

-- (TUTOR Ver) Print Scanned Map! --
-----
      ~ ~
      M
      M
      # #
-----
```

RSE1201 – C Programming

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

Returned (scanned) terrain symbol is : ''
Initial Enrg : 120000, Tot Enrg Used : 34330, Tot Shld Enrg Used : 100, Engy Left : 85670, Shld Enrg Left : 29900

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

  ~ ~
  M  M
  -
# # #

Enter your command (q to quit) : 2

Error! Scanning location is BEYOND map range!
Returned (scanned) terrain symbol is : ''
Initial Enrg : 120000, Tot Enrg Used : 34510, Tot Shld Enrg Used : 100, Engy Left : 85490, Shld Enrg Left : 29900

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

  ~ ~
  M  M
  -
# # #

Enter your command (q to quit) : 3

Error! Scanning location is BEYOND map range!
Returned (scanned) terrain symbol is : ''
Initial Enrg : 120000, Tot Enrg Used : 35230, Tot Shld Enrg Used : 100, Engy Left : 84770, Shld Enrg Left : 29900

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

  ~ ~
```

RSE1201 – C Programming

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

Returned (scanned) terrain symbol is : ''
Initial Enrg : 120000, Tot Enrg Used : 35230, Tot Shld Enrg Used : 100, Engy Left : 84770, Shld Enrg Left : 29900

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      ~ ~
      M M
      -
    # # #

Enter your command (q to quit) : t
Toggle Scan Radius => now set to 1 terrain units!

Enter your command (q to quit) : 4

Returned (scanned) terrain symbol is : 'X'
Initial Enrg : 120000, Tot Enrg Used : 35320, Tot Shld Enrg Used : 100, Engy Left : 84680, Shld Enrg Left : 29900

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      ~ ~
      M M
      X -
    # # #

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.
Initial Enrg : 120000, Tot Enrg Used : 36320, Tot Shld Enrg Used : 5100, Engy Left : 83680, Shld Enrg Left : 24900

-----
-- (TUTOR Ver) Print Explored Map! --
-----
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-----
-- (TUTOR Ver) Print Explored Map! --
-----

      M
      |
  X   |

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.
Initial Enrg : 120000, Tot Enrg Used : 36420, Tot Shld Enrg Used : 5100, Engy Left : 83580, Shld Enrg Left : 24900

-----
-- (TUTOR Ver) Print Explored Map! --
-----

      M
      |
  _  X |

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.
Initial Enrg : 120000, Tot Enrg Used : 36520, Tot Shld Enrg Used : 5100, Engy Left : 83480, Shld Enrg Left : 24900

-----
-- (TUTOR Ver) Print Explored Map! --
-----

      M
      |
  _  | X |

Enter your command (q to quit) : a
```



```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

Initial Enrg : 120000, Tot Enrg Used : 37320, Tot Shld Enrg Used : 5500, Engy Left : 82680, Shld Enrg Left : 24500

-----
-- (TUTOR Ver) Print Explored Map! --
-----

      j j  █  █  x  █  M

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.

Initial Enrg : 120000, Tot Enrg Used : 37420, Tot Shld Enrg Used : 5500, Engy Left : 82580, Shld Enrg Left : 24500

-----
-- (TUTOR Ver) Print Explored Map! --
-----

    _ j j  █  █  x  █  M

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.

Initial Enrg : 120000, Tot Enrg Used : 37520, Tot Shld Enrg Used : 5500, Engy Left : 82480, Shld Enrg Left : 24500

-----
-- (TUTOR Ver) Print Explored Map! --
-----

    _  █ j j  █  █  x  █  M
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-- (TUTOR Ver) Print Explored Map! --
-----
      _  j j  x M
-----

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.
Initial Enrg : 120000, Tot Enrg Used : 37920, Tot Shld Enrg Used : 5525, Engy Left : 82080, Shld Enrg Left : 24475

-- (TUTOR Ver) Print Explored Map! --
-----
      h  j j  x M
-----

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.
Initial Enrg : 120000, Tot Enrg Used : 38320, Tot Shld Enrg Used : 5550, Engy Left : 81680, Shld Enrg Left : 24450

-- (TUTOR Ver) Print Explored Map! --
-----
      h h  j j  x M
-----

Enter your command (q to quit) : a
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-- (TUTOR Ver) Print Explored Map! --
-----
      h h  j j  x M
-----

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.

Initial Enrg : 120000, Tot Enrg Used : 39120, Tot Shld Enrg Used : 5650, Engy Left : 80880, Shld Enrg Left : 24350

-- (TUTOR Ver) Print Explored Map! --
-----
      M h h  j j  x M
-----

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.

Initial Enrg : 120000, Tot Enrg Used : 39220, Tot Shld Enrg Used : 5650, Engy Left : 80780, Shld Enrg Left : 24350

-- (TUTOR Ver) Print Explored Map! --
-----
      M h h  j j  x M
-----
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

Initial Enrg : 120000, Tot Enrg Used : 39220, Tot Shld Enrg Used : 5650, Engy Left : 80780, Shld Enrg Left : 24350

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  _ M h h  0 0 j j  0 0 X 0 M

Enter your command (q to quit) : 2

Returned (scanned) terrain symbol is : '#'

Initial Enrg : 120000, Tot Enrg Used : 39310, Tot Shld Enrg Used : 5650, Engy Left : 80690, Shld Enrg Left : 24350

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

  _
  #

          ~ ~
          M M
          X
          # # #

Enter your command (q to quit) : 1

Returned (scanned) terrain symbol is : '#'

Initial Enrg : 120000, Tot Enrg Used : 39670, Tot Shld Enrg Used : 5650, Engy Left : 80330, Shld Enrg Left : 24350

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

  _
  #

          ~ ~
          M M
          X
          # # #
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      ~ ~
      M M
      X M
      # # #

# #

Enter your command (q to quit) : 3

Returned (scanned) terrain symbol is : '#'

Initial Enrg : 120000, Tot Enrg Used : 40030, Tot Shld Enrg Used : 5650, Engy Left : 79970, Shld Enrg Left : 24350

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      ~ ~
      M M
      X M
      # # #

# # #

Enter your command (q to quit) : 4

Returned (scanned) terrain symbol is : ' '

Initial Enrg : 120000, Tot Enrg Used : 40120, Tot Shld Enrg Used : 5650, Engy Left : 79880, Shld Enrg Left : 24350

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      ~ ~
      M M
      X M
      # # #

# # #

Enter your command (q to quit) : 6
```

RSE1201 – C Programming

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      ~ ~
      M ~
      M
X     M
# # #

Enter your command (q to quit) : 7

Returned (scanned) terrain symbol is : 'X'

Initial Enrg : 120000, Tot Enrg Used : 40570, Tot Shld Enrg Used : 5650, Engy Left : 79430, Shld Enrg Left : 24350

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      ~ ~
      M ~
      M
X     M
# # #

Enter your command (q to quit) : 8

Returned (scanned) terrain symbol is : ' '

Initial Enrg : 120000, Tot Enrg Used : 40660, Tot Shld Enrg Used : 5650, Engy Left : 79340, Shld Enrg Left : 24350

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

      ~ ~
      M ~
      M
X     M
# # #

Enter your command (q to quit) : 9
```

```
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-----
-- (TUTOR Ver) Print Scanned Map! --
-----

  X   M
  _   M
  #   #   #

          ~ ~
          M ~
          M
  X   M
  #   #   #

Enter your command (q to quit) : s

INVALID / IMPASSABLE LOCATION! Unable to move SOUTH!
Initial Enrg : 120000, Tot Enrg Used : 42020, Tot Shld Enrg Used : 10650, Engy Left : 77980, Shld Enrg Left : 19350

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  _   M   h   h   █   █   j   j   █   █   X   █   M

Enter your command (q to quit) : w

Vehicle moved 'Up / North' by 1 terrain unit.
Initial Enrg : 120000, Tot Enrg Used : 42120, Tot Shld Enrg Used : 10650, Engy Left : 77880, Shld Enrg Left : 19350

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  █   M   h   h   █   █   j   j   █   █   X   █   M

Enter your command (q to quit) : w
```

```
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Enter your command (q to quit) : w

Vehicle moved 'Up / North' by 1 terrain unit.
Initial Enrg : 120000, Tot Enrg Used : 43120, Tot Shld Enrg Used : 11650, Engy Left : 76880, Shld Enrg Left : 18350

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  E                                     M
  |                                     |
  | M h h | | j j | | x |
  |                                     |

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.
Initial Enrg : 120000, Tot Enrg Used : 43320, Tot Shld Enrg Used : 11700, Engy Left : 76680, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Explored Map! --
-----

 f E                                     M
  |                                     |
  | M h h | | j j | | x |
  |                                     |

Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.
Initial Enrg : 120000, Tot Enrg Used : 43420, Tot Shld Enrg Used : 11700, Engy Left : 76580, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Explored Map! --
-----
```

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```
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Enter your command (q to quit) : a

Vehicle moved 'Left / West' by 1 terrain unit.

Initial Enrg : 120000, Tot Enrg Used : 43420, Tot Shld Enrg Used : 11700, Engy Left : 76580, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Explored Map! --
-----

  _ f E                                     M
  | | | | | | | | | | | | | | | | | | | |
  M h h | | j j | | | | | | | | | | | |

Enter your command (q to quit) : 1

Returned (scanned) terrain symbol is : '#'

Initial Enrg : 120000, Tot Enrg Used : 43780, Tot Shld Enrg Used : 11700, Engy Left : 76220, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

# _ X M                                     ~ ~
# # # M M                                     X X
# # #                                     # # #

Enter your command (q to quit) : 4

Returned (scanned) terrain symbol is : '#'

Initial Enrg : 120000, Tot Enrg Used : 43870, Tot Shld Enrg Used : 11700, Engy Left : 76130, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#                                     ~ ~
#                                     M
```

RSE1201 – C Programming

```
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Initial Enrg : 120000, Tot Enrg Used : 43870, Tot Shld Enrg Used : 11700, Engy Left : 76130, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

# - ~ ~
# X M
# M
# # #
# # #

Enter your command (q to quit) : 7

Returned (scanned) terrain symbol is : '#'

Initial Enrg : 120000, Tot Enrg Used : 44230, Tot Shld Enrg Used : 11700, Engy Left : 75770, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

# - ~ ~
# X M
# M
# # #
# # #

Enter your command (q to quit) : t

Toggle Scan Radius => now set to 2 terrain units!

Enter your command (q to quit) : 1

Error! Scanning location is BEYOND map range!

Returned (scanned) terrain symbol is : ''

Initial Enrg : 120000, Tot Enrg Used : 44950, Tot Shld Enrg Used : 11700, Engy Left : 75050, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Scanned Map! --
-----
```

```
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-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#           ~ ~
#  -       M
#  X       M
#           M
#   #   #
#   #   #

Enter your command (q to quit) : 2

Returned (scanned) terrain symbol is : 'E'

Initial Enrg : 120000, Tot Enrg Used : 45130, Tot Shld Enrg Used : 11700, Engy Left : 74870, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#           ~ ~
#  -       M
#  X       M
#  E       M
#   #   #
#   #   #

Enter your command (q to quit) : 3

Returned (scanned) terrain symbol is : ' '

Initial Enrg : 120000, Tot Enrg Used : 45850, Tot Shld Enrg Used : 11700, Engy Left : 74150, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#           ~ ~
#  -       M
#  X       M
#  E       M
#   #   #
#   #   #

Enter your command (q to quit) : 4
```

```
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E      M      X      #
#      #      #      #

Enter your command (q to quit) : 4

Error! Scanning location is BEYOND map range!
Returned (scanned) terrain symbol is : ''
Initial Enrg : 120000, Tot Enrg Used : 46030, Tot Shld Enrg Used : 11700, Engy Left : 73970, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#      ~      ~
#      M      M
#      X      M
#      E      M
#      #      #

Enter your command (q to quit) : 6

Returned (scanned) terrain symbol is : 'E'
Initial Enrg : 120000, Tot Enrg Used : 46210, Tot Shld Enrg Used : 11700, Engy Left : 73790, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#      ~      ~
#      M      M
#      X      M
#      E      M
#      E      M
#      #      #

Enter your command (q to quit) : 7

Error! Scanning location is BEYOND map range!
Returned (scanned) terrain symbol is : ''
Initial Enrg : 120000, Tot Enrg Used : 46930, Tot Shld Enrg Used : 11700, Engy Left : 73070, Shld Enrg Left : 18300
```

RSE1201 – C Programming

```
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Error! Scanning location is BEYOND map range!
Returned (scanned) terrain symbol is : ' '
Initial Enrg : 120000, Tot Enrg Used : 46930, Tot Shld Enrg Used : 11700, Engy Left : 73070, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#           ~ ~
#  _   E   M
#  X     M
#  E     M
#  #   #
#  #   #

Enter your command (q to quit) : 8

Returned (scanned) terrain symbol is : ' '
Initial Enrg : 120000, Tot Enrg Used : 47110, Tot Shld Enrg Used : 11700, Engy Left : 72890, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#           ~ ~
#  _   E   M
#  X     M
#  E     M
#  #   #
#  #   #

Enter your command (q to quit) : 9

Returned (scanned) terrain symbol is : ' '
Initial Enrg : 120000, Tot Enrg Used : 47830, Tot Shld Enrg Used : 11700, Engy Left : 72170, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#           ~ ~
#  _   E   M
#  X     M
#  #   #
#  #   #
```

```
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Initial Enrg : 120000, Tot Enrg Used : 47830, Tot Shld Enrg Used : 11700, Engy Left : 72170, Shld Enrg Left : 18300

-----
-- (TUTOR Ver) Print Scanned Map! --
-----

#           ~ ~
#  _   E   M
#  X   M
E       M
#   #   #
#   #   #

Enter your command (q to quit) : q

-----
-- (TUTOR Ver) Exploration Report! --
-----

Total no. of scans          taken : 34
Total no. of movements (steps) taken : 18

[ Energy Report ]

Initial Energy      : 120000
Total Energy Used   : 47830
Total Shield Energy Used : 11700
Energy Left         : 72170
Shield Energy Left  : 18300

[ Terrain Types Scanned ]

Type '#' : 9
Type ' ' : 9
Type 'j' : 0
Type 'M' : 4
Type '~' : 2
Type 'X' : 2
Type 'S' : 0
Type 'E' : 2
Type '?' : 0
Type 'f' : 0
Type 'h' : 0
Type 'w' : 0

[ Terrain Types Explored ]

Type '#' : 0
Type ' ' : 9
Type 'j' : 2
Type 'M' : 2
Type '~' : 0
Type 'X' : 1
Type 'S' : 0
```

```
student@Ubuntu-22044-VBox: ~/Desktop/RSE1201/ProjectFiles/MS1_Pkg

#   X   M
E   M
#   #   #

Enter your command (q to quit) : q

-----
-- (TUTOR Ver) Exploration Report! --
-----

Total no. of scans          taken : 34
Total no. of movements (steps) taken : 18

[ Energy Report ]

Initial Energy      : 120000
Total Energy Used   : 47830
Total Shield Energy Used : 11700
Energy Left         : 72170
Shield Energy Left  : 18300

[ Terrain Types Scanned ]

Type '#' : 9
Type '.' : 9
Type 'j' : 0
Type 'M' : 4
Type '~' : 2
Type 'X' : 2
Type 'S' : 0
Type 'E' : 2
Type '?' : 0
Type 'f' : 0
Type 'h' : 0
Type 'w' : 0

[ Terrain Types Explored ]

Type '#' : 0
Type '.' : 9
Type 'j' : 2
Type 'M' : 2
Type '~' : 0
Type 'X' : 1
Type 'S' : 0
Type 'E' : 1
Type 'f' : 1
Type 'h' : 2
Type 'w' : 0

[ Mission Statistics ]

Mission Type      : Find All Danger Points
Danger Points found : 2
Completion Status  : 66.67 %
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```